Problem Set 2

Starter

1 (E)**.** Suppose that the vectors $\mathbf{e}_1, \ldots, \mathbf{e}_n$ form a basis for a real vector space V. Is $\mathbf{e}_1 + \mathbf{e}_2,\ \mathbf{e}_2 + \mathbf{e}_3, \ldots, \mathbf{e}_{n-1} + \mathbf{e}_n,\ \mathbf{e}_n$ also a basis?

2 (E)**.** Let $\{v_1, v_2, \ldots, v_n\}$ be a linearly independent set of n vectors in a real vector space V. Prove that each of the following sets is also linearly independent:

(a) $\{c_1 v_1, c_2 v_2, \ldots, c_n v_n\}$ where $c_i \neq 0$ for $1 \leq i \leq n$;

(b) $\{w_1, w_2, \ldots, w_n\}$ where $w_i = v_i + v_1$ for $1 \leq i \leq n$.

3 (E)**.** Let $V = \mathbb{R}^{\mathbb{R}} = \{f : \mathbb{R} \to \mathbb{R}\}$. Determine whether the sets $\{p, q, r\}$ is linearly independent in V, where $p(x) = \cos^2(x)$, $q(x) = \cos(2x)$ and $r(x) = 1$.

4 (M)**.** Prove the Steinitz Exchange Lemma: Let V be a vector space over a field $\mathbb{F}$. Suppose $x_1, \ldots, x_m$ are linearly independent and $y_1, \ldots, y_n$ span V. Then $m \leq n$ and the vectors $y_1, \ldots, y_n$ can be reordered so that $x_1, \ldots, x_m,\ y_{m+1}, \ldots, y_n$ span V.

Deduce that, for a subset S of $\mathbb{R}^n$, any two of the following imply the third:

(i) S is linearly independent;

(ii) S is spanning;

(iii) S has exactly n elements.

Advanced

5 (L-M)**.** Let $n \geq 2$. The vectors $a_1, a_2, \ldots, a_n$ in $\mathbb{R}^n$ have the property that every subset comprising $n-1$ of the vectors is linearly independent. Are $a_1, a_2, \ldots, a_n$ necessarily linearly independent? Justify your answer by a proof or a counterexample.

6 (L-M)**.** Given an integer $n > 2$, is it always possible to find a set of vectors $\{c_1, c_2, \ldots, c_n\}$ in $\mathbb{R}^n$ with the property that every pair is linearly independent and that every triple is linearly dependent? Justify your answer.

7 (M)**.** Let $x_1, \ldots x_m$ be column vectors of dimension n over $\mathbb{R}$, where $m > 1$. Let $y = x_1 + \cdots + x_m$. Show that $y - x_1, y - x_2, \ldots, y - x_m$ are linearly independent if and only if $x_1, \ldots, x_m$ are linearly independent.

8 (H)**.** Let V be a finite-dimensional vector space over an infinite field $\mathbb{F}$. Prove that if $S_1, \ldots, S_k$ are subspaces of V of equal dimension, then there is a subspace T of V for which $V = S_i \oplus T$ for all $i = 1, \ldots, k$. In other words, T is a common complement of the subspaces S_i.